

Taylor Tang

MACHINE LEARNING ENGINEER · CONSULTANT

Brisbane, QLD, Australia

☎ (+61) 422-110-216 | ✉ tay.tang@outlook.com | 🌐 www.github.com/TangChao729 | 🔗 www.linkedin.com/in/taylor-tang

"Better to run than curse the road"

Summary

Machine Learning Engineer with a strong focus on computer vision and deep learning, bringing over two years of experience in developing end-to-end AI solutions using satellite and UAV imagery. Skilled at building scalable ML systems, from data collection and annotation to model training, evaluation, and deployment on cloud platforms like AWS.

Proficient in Python, PyTorch, YOLO, and ViT, with hands-on expertise across object detection, image segmentation, and model optimization. Passionate about applying AI to real-world challenges and delivering high-impact, data-centric solutions. Known for a proactive mindset, rapid learning, and bridging research innovation with production-ready implementation.

Work Experience

Machine Learning Consultant

AT KJR

Brisbane, Australia

Aug. 2024 - Present

- Consulting with clients from various backgrounds, proposing AI-driven solutions and prototyping applications.
- Building commercial level cloud applications using AWS to host and deploy AI models, ensuring scalability and reliability.
- Engaging in latest AI research, monitoring and enhancing existing AI models to improve performance and adapt to new challenges.

Algorithm & AI Tutor

AT EIC EDUCATION

Remote

Jan. 2024 - Present

- Teaching algorithm and AI concepts to students, focusing on using simple language and practical examples to explain complex topics.
- Creating engaging learning materials and exercises to help students understand and apply algorithmic thinking.

Machine Learning Engineer

AS FREELANCER

Remote

Jan. 2024 - Aug. 2024

- Consulted with clients to understand their AI needs and provided tailored AI-driven solutions to enhance their business operations.
- Designed and prototyped applications using cloud technologies and chatGPT APIs, ensuring efficient and scalable deployment.

Machine Learning Engineer Intern

AT VISION HQ

Brisbane, Australia

Dec. 2022 - Dec. 2023

- Learned to train computer vision models, integrated CI/CD pipelines to streamline model development and production workflows.
- Partnered with software engineers to deploy models into production systems, ensuring robust performance and scalability.

Project Manager

AT TURNER & TOWNSEND

Brisbane, Australia

Dec. 2018 - Jul. 2021

- Delivered construction projects within scope, budget, and timelines by coordinating teams and maintaining clear communication with clients.
- Analyzed project data to identify risks and opportunities, implementing strategies to mitigate issues and enhance project outcomes.

Scuba Diving Instructor

AT TANGALOOMA

Brisbane, Australia

Jan. 2015 - Dec. 2017

- Led scuba diving tours as a certified PADI instructor, emphasizing safety and engagement.

Project Highlight

Australia Age Assurance Trial – Facial Verification System

AT KJR

Brisbane, Australia

Mar. 2025 – Jun. 2025

- Contributed to the building of the nationally backed Age Assurance Technology Trial platform and integrated 17 third-party digital ID providers.
- Provided testers (students) seamless experience on evaluating age verification methods, including facial estimation and document verification.
- Implemented custom redirect flows, WebSocket-based real-time result delivery, and secure token/session handling across diverse vendor APIs.
- **Tech Stack:** React, TypeScript, Node.js (Express), PostgreSQL, AWS

Feral Animal Detection System – UAV Thermal Imagery

AT KJR

Brisbane, Australia

Aug. 2024 – Mar. 2025

- Led end-to-end development of a computer vision pipeline using UAV-collected thermal footage for local government wildlife management.
- Processed and annotated raw UAV thermal imagery; managed experiments and training CV models, boosting detection accuracy from 59% to 82% mAP@0.5 through successive data curation and model tuning iterations.
- Designed a flexible hierarchical label schema, enabling iterative development and consistent performance tracking across dataset versions.
- Delivered actionable insights to stakeholders on feral animal distribution, supporting targeted pest control and conservation efforts.
- **Tech Stack:** Python, YOLOv8, CVAT, FiftyOne, ClearML, UAV thermal imagery

Road safety monitoring system with dash-cam imagery

AT VISION HQ

Brisbane, Australia

Jan. 2023 - Dec. 2023

- Contributed to developing CV workflow using vehicle dash-cam data and YOLOv5 model to detect road hazards like potholes and debris.
- Provided clients with precise hazard details for faster and more accurate urban infrastructure management.
- **Tech Stack:** Python, YOLOv5, OpenCV, PyTorch

National Bushfire Analysis Using Satellite Imagery

AT THE UNIVERSITY OF MELBOURNE

Melbourne, Australia

Aug. 2023 – Nov. 2023

- Conducted a comparative study on burnt area detection techniques to support Australia’s National Bushfire Management Policy, and delivered research-backed insights for bushfire recovery and planning.
- Benchmarked traditional burnt area assessment with classical CNN models and emerging Transformer-based approaches.
- Built a custom dataset from scratch by collecting, cleaning, and annotating multi-spectral satellite imagery from earth observation platforms.
- Enhanced the SegFormer model to process multi-spectral data, achieving F1-score of 88%, outperforming its RGB-only baseline by 3%.
- Research accepted and presented at the IEEE eScience 2024 conference.
- **Tech Stack:** Python, PyTorch, SegFormer, Satellite APIs, Remote Sensing, Multi Spectral Image Preprocessing

Honors & Awards

2023	Dean’s Honours List , The University of Melbourne	Melbourne, Australia
2022	Dean’s Honours List , The University of Melbourne	Melbourne, Australia
2022	Scholarship , The University of Melbourne	Melbourne, Australia
2022	2nd place , Dota2 - Australia Esports League University	Melbourne, Australia
2019	Above & Beyond New Employee , Turner & Townsend	Brisbane, Australia
2019	Community Volunteer , Yalari Education Event	Brisbane, Australia
2019	Community Volunteer , Yalari Indigenous Children	Brisbane, Australia
2018	Vice Chancellor’s List , Bond University	Gold Coast, Australia
2018	Dean’s Honours List , Bond University	Gold Coast, Australia
2017	Vice Chancellor’s List , Bond University	Gold Coast, Australia
2017	Dean’s Honours List , Bond University	Gold Coast, Australia

Certificates

2024	Foundation Tester , International Software Testing Qualifications Board (ISTQB)
2024	Developer Associate , Azure
2024	AI Engineer Associate , Azure
2023	Cloud Practitioner , Amazon Web Services (AWS)

Education

The University of Melbourne

MASTER OF IT (AI MAJOR) AT FACULTY OF ENGINEERING AND INFORMATION TECHNOLOGY (FEIT)

Melbourne, Australia

Jan. 2022 - Nov. 2023

- WAM: 83/100, Graduate with Distinction

Bond University

MASTER OF CONSTRUCTION PRACTICE

Gold Coast, Australia

Jan. 2017 - Nov. 2018

- GPA: 6.7/7.0, Graduate with Distinction

Griffith University

BACHELOR OF BUSINESS MANAGEMENT (TOURISM & HOSPITALITY MAJOR)

Brisbane, Australia

Oct. 2012 - Nov. 2014